Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

Parallel branches draw 0.3 A and 0.5 A. Find the current supplied to their junction.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $P = IV$; $E = P\Delta t$; $P = I^2R = V^2/R$.
Constant power for $E = P\Delta t$; last equalities apply to an ohmic resistor.

Worked reasoning: A device at 6 V drawing 2 A receives power VI=12 W. Over 5 s it receives 60 J. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and label voltage, current, power and elapsed time with distinct units.

Marzano check - circle: ① ② ③

03 Look and explain

Two motors use the same energy but one takes half the time. Compare average power.

Marzano check - circle: ① ② ③

04 Build the solution

A motor draws 2 A at 6 V. Find P=2×6, then energy used in 5 s using E=P×5.

Marzano check - circle: ① ② ③

05 Use the idea

A student says a 10 W motor always uses more energy than a 5 W motor. Supply a time comparison that disproves this.

Marzano check - circle: ① ② ③

06 Choose a method

A wave travels at fixed speed. A second target is 4 times farther away than the first. Compare one-way travel times.

Marzano check - circle: ① ② ③

07 Exit check

A 3 W device runs for 10 s. Find energy.

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

Two devices use the same 60 J: one runs at 10 W, another at 20 W. Find their run times.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.